NOTE

To get the equation of the tangent at (x_1, y_1) to the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \ldots (1)$$

replace x^2 by xx_1, y^2 by yy_1, x by $\frac{x + x_1}{2}$ and y by $\frac{y + y_1}{2}$ in equation (1).

Example 3.27

Find the equation of tangent at the point (–2, 5) on the circle

$$x^2 + y^2 + 3x - 8y + 17 = 0.$$

Solution

The equation of the tangent at (x_1, y_1) to the given circle $x^2 + y^2 + 3x - 8y + 17 = 0$ is

$$xx_1 + yy_1 + 3 \times \frac{1}{2}(x + x_1) - 8 \times \frac{1}{2}(y + y_1) + 17 = 0$$

Here $(x_1, y_1) = (-2, 5)$

$$-2x + 5y + \frac{3}{2}(x - 2) - 4(y + 5) + 17 = 0$$

$$-2x + 5y + \frac{3}{2}x - 3 - 4y - 20 + 17 = 0$$

$$-4x + 10y + 3x - 6 - 8y - 40 + 34 = 0$$

$x - 2y + 12 = 0$ is the required equation.

Length of the tangent to the circle

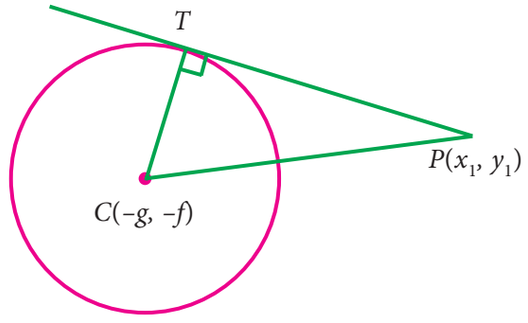


Fig. 3.8

Length of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ from a point $P(x_1, y_1)$ is PT = $\sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$

NOTE

(i) If the point P is on the circle then $PT^2 = 0$

(ii) If the point P is outside the circle then $PT^2 > 0$

(iii) If the point P is inside the circle then $PT^2 < 0$

Example 3.28

Find the length of the tangent from the point (2, 3) to the circle

$$x^2 + y^2 + 8x + 4y + 8 = 0$$

Solution

The length of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ from a point (x_1, y_1) is $\sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$

Length of the tangent

$$= \sqrt{x_1^2 + y_1^2 + 8x_1 + 4y_1 + 8}$$

$$= \sqrt{2^2 + 3^2 + 8(2) + 4(3) + 8}$$

[Here $(x_1, y_1) = (2, 3)$]

$$= \sqrt{49}$$

Length of the tangent = 7 units

Example 3.29

Determine whether the points $P(0,1)$, $Q(5,9)$, $R(-2, 3)$ and $S(2, 2)$ lie outside the circle, on the circle or inside the circle $x^2 + y^2 - 4x + 4y - 8 = 0$

Solution

The equation of the circle is $x^2 + y^2 - 4x + 4y - 8 = 0$

$$PT^2 = x_1^2 + y_1^2 - 4x_1 + 4y_1 - 8$$

At $P(0, 1)$

$PT^2 = 0 + 1 + 0 + 4 - 8 = -3 < 0$

At $Q(5, 9)$

$QT^2 = 25 + 81 - 20 + 36 - 8 = 114 > 0$

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